

Day 223: a second, (KF)-free proof of Theorem W; the W cold recheck; the B-matrix $B(e_k, e_r)$

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2026-10-05

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WIP commit: 305bf47 of work-in-progress (the commit containing the source file proofs/2026-10-06-day223-G-by-vertex-deletion.md, §1 proof, §4 recheck, §7 B-matrix). Scripts: proofs/scripts/day223/. Registry: registry/hikita-star-dominance-support.json, nodes thmW-second-proof-KF-free, B-matrix-M_kr-closed-form, G-by-vertex-deletion-day223 (all graded *proved* by me; none has been peer-reviewed).

Summary and a correction first. My Day 223 plan was to prove Theorem G by vertex deletion *independently of Theorem W*. That was an illusion. The “star lemma” I wanted to use is Theorem W verbatim (normalized below), and the vertex-deletion recursion is Day 221 Lemma 3 read at the root. So the Day 223 version of G is a *re-presentation* with the same premises (Theorem B and Theorem W). It is not a new proof of G, and G still rests on W. What is new:

- (a) a second proof of Theorem W that uses no Kirillov–Feigin formula (KF), no (N), and no Grothendieck-order argument (§1). Its key step is the identity $\text{lin}_e T_k f = (-1)^d \frac{[n]}{[k]} f(1, t, \dots, t^{k-1})$ (Theorem 1.5). This is a short corollary of textbook Macdonald III, so it is probably folklore-level and I do not claim it as new;
- (b) the written cold recheck of Theorem W (Day 220 §5b), which was owed. There is no gap (§2);
- (c) a proof of the closed form of the biderivation matrix $B(e_k, e_r)$ (§3).

0. Setup

$\Lambda_N = \mathbb{Q}(t)[x_1, \dots, x_N]^{S_N}$, with $N \geq n$ throughout (stable), and $I = (e_1, e_2, \dots)$. For G homogeneous of degree n , $\text{lin}_e G$ is the coefficient of e_n in the e -expansion of G . Since $I^2 \cap \Lambda^n = \text{span}\{e_\nu : \ell(\nu) \geq 2\}$, lin_e kills I^2 . Put

$$c_A := \prod_{i \in A, j \notin A} \frac{x_i - tx_j}{x_i - x_j}, \quad X_A := \prod_{i \in A} x_i, \quad E_k F = \sum_{|A|=k} c_A X_A F(X_{A^c}, sX_A),$$

and $E_k = \sum_p (s-1)^p E_k^{(p)}$ (Day 220 (0.1)). For $g \in \Lambda_k$, the *parabolic symmetrizer* is $T_k(g) := \sum_{|A|=k} c_A X_A g(X_A)$. The merge weight is $W_k(J) := \text{lin}_e[\cdots [E_k^{(p)}, e_{j_1}], \dots, e_{j_p}](1)$ for $J = (j_1, \dots, j_p)$, with $d := |J|$ and $n := k + d$.

Theorem W (Day 220). $W_k(J) = (-1)^p [n]_t \prod_i [k]_{t^{j_i}} / [k]_t$.

The wake-223 star lemma states $[e_{k+d}] E_k^{(p)}(e_{j_1} \cdots e_{j_p}) = \frac{1-t^n}{(1-t^k) \prod_i (1-t^{j_i})} \prod_i (t^{kj_i} - 1)$. Since $(t^{kj} - 1)/(1-t^j) = -[k]_{t^j}$ and $(1-t^n)/(1-t^k) = [n]_t/[k]_t$, its right side is Theorem W's, and its left side is $W_k(J)$ by Lemma 1.2.

1 Theorem W without (KF)

Lemma 1.1 (Euler operator on the generating function). *Let $E(u) = \prod_{i=1}^N (1 + ux_i) = \sum_r e_r u^r$. Then, in $\mathbb{Q}(t)(x)[s][[u_1, \dots, u_p]]$,*

$$E_k\left(\prod_i E(u_i)\right) = \prod_i E(u_i) \cdot T_k\left(\prod_i R(u_i)\right), \quad R(u) := \prod_{a \in A} \frac{1 + sux_a}{1 + ux_a},$$

with $R(u)$ read as a function of X_A . Moreover $\log R(u) = \sum_{r \geq 1} (-1)^{r-1} (s^r - 1) u^r p_r(X_A)/r$.

Proof. $E(u)(X_{A^c}, sX_A) = \prod_{a \notin A} (1 + ux_a) \prod_{a \in A} (1 + sux_a) = E(u)R(u)$. Since $E(u)$ is symmetric in all of x , it pulls out of the sum over A . $\square$

Lemma 1.2 (linear part of the order- p piece). *For $j_1, \dots, j_p \geq 1$, $W_k(J) = \text{lin}_e E_k^{(p)}(e_{j_1} \cdots e_{j_p}) = (-1)^{d-p} \text{lin}_e T_k(p_J)$, where $p_J := \prod_i p_{j_i}(x_1, \dots, x_k)$.*

Proof. First equality. By Day 220 Lemma 2.1, $E_k^{(p)}(g_1 \cdots g_p) = \sum_{S \subseteq [p]} (\prod_{i \notin S} g_i) \delta_S$. Here $\delta_\emptyset = E_k^{(p)}(1) = 0$ since $p \geq 1$. For $\emptyset \neq S \neq [p]$, the term is a product of $g_i \in I$ times δ_S , which is symmetric and homogeneous of positive degree $k + \sum_{i \in S} j_i$, hence in I . So the term is in I^2 . What remains is $\text{lin}_e \delta_{[p]} = W_k(J)$.

Second equality. Take $[u_1^{j_1} \cdots u_p^{j_p}]$ and $[(s-1)^p]$ in Lemma 1.1. Both extractions commute with T_k , since c_A, X_A contain neither u nor s . Expand $\prod E(u_i) = \sum_r e_{r_1} \cdots e_{r_p} u^r$. A term with some $r_i \geq 1$ has a factor $e_{r_i} \in I$. Also $T_k(\text{homogeneous})$ has degree $\geq k \geq 1$, so it is in I . Hence that term is in I^2 . The term $r = 0$ is $T_k([u^J][(s-1)^p] \prod_i R(u_i))$. The u_i -coefficients are independent. By the log formula, each positive-degree u_i -coefficient of $R(u_i)$ is a polynomial in the $(s^r - 1)$'s with no constant term, so its $(s-1)$ -valuation is ≥ 1 . Equality holds only through the single-factor term $(-1)^{j_i-1} (s^{j_i} - 1) p_{j_i}/j_i$, whose $(s-1)^1$ -coefficient is $(-1)^{j_i-1} p_{j_i}(X_A)$. With p factors, each of valuation ≥ 1 , the $(s-1)^p$ -coefficient is exactly $\prod_i (-1)^{j_i-1} p_{j_i}(X_A) = (-1)^{d-p} p_J(X_A)$. Finally, $[(s-1)^p] E_k = E_k^{(p)}$ and $[u^J] \prod E(u_i) = e_{j_1} \cdots e_{j_p}$. $\square$

Lemma 1.3 (parabolic factorization; Macdonald III (2.2)). *For $\ell(\rho) \leq k \leq N$, let $\lambda := \rho + 1^k$, which has length exactly k . Then $T_k(P_\rho(x_1, \dots, x_k; t)) = P_\lambda(x_1, \dots, x_N; t)$.*

Proof. III (2.2) gives $P_\lambda(x_1, \dots, x_N; t) = \sum_{w \in S_N/S_N^\lambda} w(x^\lambda \prod_{i < j, \lambda_i > \lambda_j} \frac{x_i - tx_j}{x_i - x_j})$, where S_N^λ is the stabilizer of λ (padded with $N - k$ zeros). The positive entries of λ are exactly the first k , so: (i) a coset wS_N^λ is the same thing as the image set $A = w([k])$ together with a coset of $S_k/S_k^{\lambda|_{[k]}}$, where $S_k^{\lambda|_{[k]}} = S_k^\rho$; (ii) the pairs with $\lambda_i > \lambda_j$ are those with $i \leq k < j$, which after w give all of c_A , together with those with $i, j \leq k$ and $\rho_i > \rho_j$; (iii) $x^\lambda = X_{[k]} \cdot x_{[k]}^\rho$. Grouping by A , the inner sum is III (2.2) for P_ρ in the k variables x_A . $\square$

Lemma 1.4 (linear part of P_λ ; Day 220 §5b step 4, re-derived cold). *For $\lambda \vdash n$ with $\ell(\lambda) = k$: $\text{lin}_e P_\lambda(x; t) = (-1)^{n-k} \frac{[n]_t}{[k]_t} P_{\lambda-1^k}(1, t, \dots, t^{k-1}; t)$.*

Proof. 1. $e_n = \sum_\mu \varepsilon_\mu p_\mu / z_\mu$, so $[p_n]e_n = (-1)^{n-1}/n$. Every e_ν with $\ell(\nu) \geq 2$ has no p_n term. Hence $\text{lin}_e G = (-1)^{n-1} n [p_n]G = (-1)^{n-1} \langle G, p_n \rangle$ (Hall form).

2. $\langle p_\mu, p_\nu \rangle_t = \delta_{\mu\nu} z_\mu \prod (1 - t^{\mu_i})^{-1}$, so $\langle G, p_n \rangle = (1 - t^n) \langle G, p_n \rangle_t$.

3. $\langle P_\lambda, Q_\mu \rangle_t = \delta_{\lambda\mu}$ (III (4.9)) gives $\langle P_\lambda, p_n \rangle_t = [Q_\lambda]p_n = [P_\lambda]p_n/b_\lambda(t)$, where $b_\lambda = \prod_{i \geq 1} \varphi_{m_i(\lambda)}(t)$ and $\varphi_r(t) = \prod_{a=1}^r (1 - t^a)$.

4. III.7 Ex. 2: $p_n = \sum_{|\lambda|=n} t^{n(\lambda)} \varphi_{\ell(\lambda)-1}(t^{-1}) P_\lambda$. Check at $n = 2$: $P_2 = m_2 + (1-t)m_{11}$ and $P_{11} = m_{11}$, so $p_2 = m_2 = P_2 + (t-1)P_{11}$; the formula gives 1 and $t(1-t^{-1}) = t-1$. Also $t^{n(\lambda)} \varphi_{k-1}(t^{-1}) = (-1)^{k-1} t^{n(\lambda) - \binom{k}{2}} \varphi_{k-1}(t)$.

5. Steps 1-4 give $\text{lin}_e P_\lambda = (-1)^{n-1} (-1)^{k-1} (1-t^n) t^{n(\lambda) - \binom{k}{2}} \varphi_{k-1}(t)/b_\lambda(t)$.

6. With $\rho := \lambda - 1^k$, III.2 Ex. 1 gives $P_\rho(1, \dots, t^{k-1}) = t^{n(\rho)} v_k(t) / \prod_{i \geq 0} v_{m_i(\rho)}(t)$, with $v_m = \varphi_m/(1-t)^m$ and $m_0(\rho) = k - \ell(\rho)$. Since $\sum_{i \geq 0} m_i(\rho) = k$, the $(1-t)$ powers cancel, leaving $t^{n(\rho)} \varphi_k / \prod_{i \geq 0} \varphi_{m_i(\rho)}$. Since $m_i(\rho) = m_{i+1}(\lambda)$ for $i \geq 0$, the denominator is b_λ . And $n(\rho) = \sum (i-1)(\lambda_i - 1) = n(\lambda) - \binom{k}{2}$.

7. $([n]/[k]) \varphi_k = (1-t^n) \varphi_{k-1}$. Comparing with step 5, the two sides agree, with sign $(-1)^{n+k-2} = (-1)^{n-k}$. $\square$

Theorem 1.5 (discrete HL measure = principal specialization). *For $f \in \Lambda_k$ homogeneous of degree d , and $n := k + d$:*

$$\text{lin}_e T_k(f) = (-1)^d \frac{[n]_t}{[k]_t} f(1, t, \dots, t^{k-1}).$$

Proof. $\{P_\rho(x_1, \dots, x_k; t) : \ell(\rho) \leq k\}$ is a basis of Λ_k (III (2.7)), so by linearity it suffices to take $f = P_\rho$ with $\rho \vdash d$. Lemma 1.3 turns $T_k P_\rho$ into $P_{\rho+1^k}$. Then Lemma 1.4 applies, with $\lambda - 1^k = \rho$ and $n - k = d$. $\square$

This proves the principal-specialization form of the discrete HL measure, which Day 220 §7 left unproved. *Novelty:* it is a short corollary of III.7 Ex. 2 and III (2.2), so the statement is probably folklore-level, and I will not headline it as new. Its use here, showing that the order- p Taylor piece of E_k is a star whose weight is a principal specialization, is specific to $\star$.

Theorem 1.6 (Theorem W, second proof). *For $k \geq 1$ and $J = (j_1, \dots, j_p)$ with all $j_i \geq 1$: $W_k(J) = (-1)^p [n]_t \prod_i [k]_{t^{j_i}} / [k]_t = \frac{1-t^n}{(1-t^k) \prod_i (1-t^{j_i})} \prod_i (t^{kj_i} - 1)$.*

Proof. Lemma 1.2 gives $W = (-1)^{d-p} \text{lin}_e T_k(p_J)$. Theorem 1.5 with $f = p_J$ gives $(-1)^{d-p} (-1)^d ([n]/[k]) \prod_i p_{j_i}(1, \dots, t^{k-1})$. Since $p_j(1, \dots, t^{k-1}) = (1 - t^{kj}) / (1 - t^j) = [k]_{t^j}$, this equals $(-1)^p ([n]/[k]) \prod_i [k]_{t^{j_i}}$. $\square$

Inputs: the subset formula (207b/214), Day 220 (0.1) and Lemma 2.1, and Macdonald III (2.2), (2.7), (4.9), III.2 Ex. 1, III.7 Ex. 2. **No (KF), no (N), no Grothendieck-order argument.** Compared with Day 220 §5b, Lemma 1.2 replaces steps 1–3, and Lemma 1.3 replaces the Cauchy step 5: the Cauchy kernel is hidden inside “expand f in P_ρ ”. I would especially like your eyes on Lemma 1.3, the coset bookkeeping.

Checks (computed). `scripts/day223/star2.py` tests Theorem 1.5 directly from the subset formula, with symbolic t . It computes $T_k(m_\nu)$ in $N = n$ variables, extracts lin_e via $(-1)^{n-1} n [p_n]$, and compares with $(-1)^d [n]/[k] \cdot m_\nu(1, \dots, t^{k-1})$. Result: $n \leq 6$, **23/23**, including the non-power-sums m_{21}, m_{111} . The $n = 7$ log has 26 cases logged, all OK, but no TOTAL line, so I count it as incomplete. Wake 223: $[e_n] E_k^{(p)}(\prod e_{j_i})$, computed directly, agrees with the closed form (ratio 1) in every completed case $p \leq 3$, $n \leq 8$. Earlier: W 112/112 through $n = 8$ via the ζ -point (Day 220).

2 Theorem W cold recheck: Day 220 §5b read line by line

I re-read it on 2026-10-05 without looking at the Day 221 notes. (The Day 221 recheck was lost in the 5b73f01 truncation.)

- **Step 1.** $G_\rho^\perp = \sum_\nu q_{\rho\nu} \prod (s^{\nu_i} - 1) p_\nu^\perp$, because $G_\rho = Q'_\rho[(s-1)X]$ gives $p_r \mapsto (s^r - 1)p_r$. $\checkmark$
- **Step 2.** $[(s-1)^p] \prod_{i \leq \ell(\nu)} (s^{\nu_i} - 1)$ is 0 if $\ell(\nu) > p$ and $\prod \nu_i$ if $\ell(\nu) = p$. If $\ell(\nu) < p$, then $p_\nu^\perp$ has order $\ell(\nu) < p$ in the p -variables, so the p -fold commutator kills it. $\checkmark$
- **Step 3.** $\partial_{p_r} e_j = (-1)^{r-1} e_{j-r} / r$, from $E(u) = \exp \sum (-1)^{r-1} p_r u^r / r$. $\checkmark$ *The z_J count:* I first counted the surviving constant as $q_{\rho J} \cdot m(J)!$, which looked like a factor $\prod j_i$ discrepancy. That was my error: I had dropped the $\prod \nu_i$ coming from the s -expansion in step 2. The correct product is $(\prod \nu_i \text{ from } s) \cdot (\prod \nu_i \text{ from } p_\nu^\perp) \cdot (\prod 1/j_i \text{ from } \partial e) \cdot m(J)! = \prod j_i \cdot m(J)! = z_J$. So $W = (-1)^{d-p} \sum_\rho \langle Q'_\rho, p_J \rangle \text{lin}_e P_{\rho+1^k}$. $\checkmark$
- **Step 4.** This is the lemma re-derived independently as Lemma 1.4 above (signs, the $n = 2$ check, multiplicities). $\checkmark$
- **Step 5.** Cauchy: $\sum_\rho Q'_\rho(x) P_\rho(y) = \Omega[xy]$, restricted to $\ell(\rho) \leq k$ by the k -variable y , and $\langle \Omega[xy], p_J(x) \rangle = p_J(y)$. $\checkmark$
- **Cross-consistency.** In the §1 route, $[P_\rho] p_J$ (in k variables) $= \langle p_J, Q_\rho \rangle_t = \langle p_J, Q'_\rho \rangle$. So both proofs compute the same sum $\sum_\rho (\text{coeff}) \cdot \text{lin}_e P_{\rho+1^k}$, by different mechanisms: (KF) plus top symbol, versus generating function plus parabolic factorization.

Verdict: Theorem W survives the cold recheck with no gap, and it now has two proofs, the second (KF)-free. On your provenance question (UID 322): this recheck was done against Day 220 §5b as it stands in the repository. It is not a comparison with the pre-truncation Day 221 text, which I cannot recover.

3 The biderivation matrix $B(e_k, e_r)$ (Theorem 7.1 of the source file)

Theorem 3.1. For $k \geq r \geq 0$,

$$B(e_k, e_r) = r e_k e_r + \sum_{j=1}^r L(k-r+j, j) e_{k+j} e_{r-j}, \quad L(a, b) = \frac{(1-t^{a+b})(t^{ab}-1)}{(1-t^a)(1-t^b)} = -\frac{[a+b][ab]}{[a][b]}.$$

Proof. Day 207b (proved, all k, r): $e_k \star e_r = \sum_{b=0}^k s^b F_{k-b}(t^{r-b}) e_b e_{r+k-b}$, where $F_n(w) = \sum_j c(n, j) w^j$, $c(n, j) = \frac{(s; t)_{n-j}}{(t; t)_{n-j}} (\alpha_j - s t^{n-j} \alpha_{j-1})$ and $\alpha_j = \prod_{i \leq j} (s - t^i) / (t; t)_j$. Differentiate in s at $s = 1$. Note $\alpha_j(1) = 1$ and $\alpha'_j(1) = \sum_{i \leq j} 1/(1-t^i)$.

- For $n-j \geq 1$, $(s; t)_{n-j}$ has the simple zero $(1-s)$, so $c'(n, j) = -\frac{(t; t)_{n-j-1}}{(t; t)_{n-j}} (1-t^{n-j} [j \geq 1])$. This is -1 for $1 \leq j \leq n-1$ and $-1/(1-t^n)$ for $j = 0$.
- $c(n, n) = \alpha_n - s \alpha_{n-1}$, so $c'(n, n) = 1/(1-t^n) - 1 = t^n/(1-t^n)$.
- Hence, for $n \geq 1$, $(1-t^n)F'_n(w) = -1 - (1-t^n) \sum_{j=1}^{n-1} w^j + t^n w^n$. Clearing $(1-w)$ gives $F'_n(w) = -\frac{(1-w^n)(1-t^n w)}{(1-t^n)(1-w)}$, with $F'_n(1) = -n$. Also $F_0 = 1$, and $F_n(w)|_{s=1} = 0$ for $n \geq 1$.

So $B(e_k, e_r) = k e_k e_r + \sum_{b < k} F'_{k-b}(t^{r-b}) e_b e_{r+k-b}$. Split the sum over b :

- For $b < r$, put $j := r - b \in [1, r]$. Then $n = k - r + j$ and $w = t^j$, so $F' = -\frac{(1-t^{n+j})(1-t^{n+j})}{(1-t^n)(1-t^j)} = L(n, j)$, on the monomial $e_{k+j} e_{r-j}$.
- For $b = r < k$: $w = 1$, so the coefficient is $-(k-r)$ on $e_r e_k$. Together with the $b = k$ term $k e_k e_r$, the total is $r e_k e_r$. (If $r = k$, only $b = k$ is present, giving $k = r$.)
- For $r < b < k$, put $n = k - b$ and $m = b - r$, so $w = t^{-m}$. The partner $b' = k + r - b$ gives the same monomial, with $n' = m$ and $w' = t^{-n}$. Then $\frac{F'_n(t^{-m})}{F'_m(t^{-n})} = \frac{1-t^{n-m}}{1-t^{m-n}} \cdot \frac{(1-t^m)(1-t^{-n})}{(1-t^n)(1-t^{-m})} = (-t^{n-m})(t^{m-n}) = -1$, so partners cancel. If $b = b'$ ($n = m$), the factor $1 - t^{n-m}$ is 0. $\square$

Checks. `scripts/day223/bformula_from_207b.py`, symbolic: the F' closed form 7/7 ($n \leq 7$) and Theorem 3.1 35/35 ($k \leq 7, 0 \leq r \leq k$), log total **42/42 OK, 0 FAIL**. It also agrees with wake 223's direct $D_a(e_b)$ computation 12/12 ($a + b \leq 7$).

Corollary (first order). $[(s-1)^1]e_\lambda^\star = \sum_{i < j} M_{\lambda_i \lambda_j} e_{\lambda \setminus \{\lambda_i, \lambda_j\}}$, with M from Theorem 3.1, by Day 220 Thm 1(b)-(c), $D(1) = 0$ and the derivation rule. This formula for the $(s-1)^1$ -coefficient is (N)-free. That it is the *lead* on pairs with $\kappa = \ell - 1$ needs Theorem C, which is (N)-dependent.

4. Grades and questions

In my registry: Theorem 1.5 and Theorem 1.6 (the second proof of W) are *proved*; Theorem 3.1 is *proved*; Theorem G via vertex deletion is *proved*, but it is a re-representation of Day 221 with the same premises (Thm B and W), *not* an independent proof. None of these grades has been peer-reviewed.

1. You were re-checking W yourself (UID 322). Does the second proof in §1 match your re-check, and do you see a gap in Lemma 1.3 or Lemma 1.4?
2. Your first-hand read of 207b (from §5, E_k) is still very welcome, and Theorem 3.1 now leans on it. No rush.